THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 2

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Two tall plants produce 75 tall and 25 dwarf plants. The parents were most likely 1

(a) both Tt **(b)** both TT **(c)** TT and Tt **(d)** Tt and tt

2. Anaerobic respiration releases energy in the: 1

(a) cytoplasm **(b)** chloroplast **(c)** ribosome **(d)** mitochondria

3. If the tip of a shoot is removed, phototropic bending: 1

(a) occurs more **(b)** reverses **(c)** increases **(d)** does not occur

4. A flower has only stamens. It is 1

(a) female **(b)** unisexual (male) **(c)** bisexual **(d)** sterile

5. A couple already has two daughters. The probability that their third child is a boy is 1

(a) 1/2 **(b)** 1/4 **(c)** 1 **(d)** 0

6. Normal blood pressure (systolic/diastolic) in human beings is: 1

(a) 120/80 mm Hg **(b)** 140/90 mm Hg **(c)** 100/100 mm Hg **(d)** 80/120 mm Hg

7. A child of age 12 is very short but healthy. The hormone lacking: 1

(a) growth hormone **(b)** insulin **(c)** adrenaline **(d)** thyroxine only

8. The zygote in humans has 1

(a) 12 chromosomes **(b)** 46 chromosomes **(c)** 23 chromosomes **(d)** 92 chromosomes

9. A bacterium reproducing asexually will show variation because of 1

(a) minor DNA copying errors **(b)** meiosis in parents

(c) crossing of two parents **(d)** gamete fusion

10. Stomata are open during the day because: 1

(a) light is absent **(b)** leaves fall

(c) guard cells take up water and swell **(d)** guard cells lose water

11. A person touches a hot plate and quickly pulls the hand. The effector is: 1

(a) brain **(b)** skin **(c)** spinal cord **(d)** muscle

12. Which statement about variation is correct? 1

(a) Only sexual reproduction ever has variation
(b) DNA copying is always perfect
(c) Inaccurate DNA copying produces variation
(d) Variation harms every species

13. Out of 160 F2 plants in a monohybrid cross, the expected number of dwarf plants is 1

(a) 40 **(b)** 20 **(c)** 80 **(d)** 120

14. Compared with anaerobic respiration, aerobic respiration releases: 1

(a) much more energy per molecule of glucose **(b)** much less energy per molecule of glucose
(c) the same energy **(d)** no energy

15. Which hormone is responsible for the wilting of leaves? 1

(a) gibberellin **(b)** cytokinin **(c)** abscisic acid **(d)** auxin

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why is impulse transmission one-directional in a synapse? 2

17. Why are variations not seen to a large extent in organisms reproducing asexually? 2

18. If a plant with genotype Rr produces 400 seeds by self-fertilisation, how many wrinkled seeds are expected? 2

19. What is the role of the lymph in the body? 2

20. Why do we need hormones in plants for growth? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Calculate the number of seeds of each phenotype expected in the F2 of a dihybrid cross, if 800 seeds are obtained. 3

22. What happens in the kidney if a person has a kidney failure? Mention treatment. 3

23. Why are the effects of hormones slower than nerve impulses? 3

24. What are the different ways in which asexual reproduction may help plants in agriculture? 3

OR

What would the F1 look like if Mendel had crossed a pure round yellow plant with a pure wrinkled green plant, according to blending theory? Why was blending theory rejected? 3

25. Why is the stomach wall not digested by its own acid? 3

OR

Explain how information is passed between two neurons. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Compare the human digestive system of herbivores and carnivores. 5

OR

Explain how the blood sugar level is controlled and what happens in diabetes. 5

27. Explain the structure and functions of the human female reproductive system, and how it supports the development of a baby. 5

OR

Explain the terms: genotype, phenotype, dominant, recessive and allele, with the help of the cross Tt x Tt. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND